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United States Patent Application Publication 20250277495
Kind Code A1
Publication Date September 04, 2025
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COMFORT FAN PROVIDING CONTINUOUS RADIAL AIR FLOW

Abstract

A fan device provides airflow throughout a radial region from the location of the fan so that people positioned around the fan simultaneously receive a continuous flow of air at a calculated velocity that provides a desired cooling or heating effect and thermal comfort, and that can be further adjusted by the operator/user.

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Family ID: **96881063**

Appl. No.: **19/066509**

Filed: **February 28, 2025**

Related U.S. Application Data

us-provisional-application US 63561082 20240304

Publication Classification

Int. Cl.: **F04D29/54** (20060101)

U.S. Cl.:

CPC **F04D29/547** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/561,082 filed Mar. 4, 2024, the entirety of which is incorporated by reference.

FIELD OF THE INVENTION

[0002] The present invention relates generally to fan devices for providing an air flow for comfort, either for cooling or heating, and, more particularly, relates to a stationary, variable speed, electrically powered, comfort-cooling and heating fan that provides a continuous radial air flow to provide thermal comfort.

BACKGROUND OF THE INVENTION

[0003] People live is wide range of climates which can range from extremes of hot and cold. Of course, in extreme environments, humans need substantial heating or cooling machinery to be comfortable. This can involve actively heating or cooling circulating air within a structure. In more moderate climate conditions, however, the simple use of a fan, or a fan with a heat source, can provide sufficient comfort. However, fans tend to be either fixed, such as ceiling fans, or portable. Portable fan devices blow air in one direction, providing comfort only for those in the path of the air blowing from the fan. To improve the distribution of air blown by a fan, oscillating mechanisms were developed so that a fan will automatically oscillate in a range of directions. This provides a moment of comfort as the fan oscillates past a given location. Once it passes that location, however, no more air is blown in that direction from the fan until the fan reverses direction and again passes by location.

[0004] It is not uncommon for people to share a table, and sit around the table. If an air flow is desired to provide comfort to such a group of people, then an oscillating fan is not an ideal solution. If the oscillating fan is placed to the side of the table, then some people will be closer to the fan than others, and those closest to the fan, as well as those farthest from the fan may receive too much, or too little air flow. If an oscillating fan is placed in the middle, then again, each person will only get a brief flow of air as the fan oscillates between angles.

[0005] Therefore, a need exists to overcome the problems with the prior art as discussed above.

SUMMARY OF THE INVENTION

[0006] The invention provides a comfort fan or fan device that overcomes the herein afore-mentioned disadvantages of the heretofore-known devices and methods of this general type and that generates a radial, continuous airflow around the circumference of the comfort fan, or a substantial portion of the circumference of the comfort fan. The rate of the air flow out of the comfort fan is selected so that the incident air flow velocity on a person within a given distance from the comfort fan, will experience a desired relief from the ambient environmental conditions. This can be either a cooling or a warming effect. The initial air flow rate can be based on industry standards that describe the comfort effect of air flow for most people.

[0007] In accordance with the inventive embodiments described herein, there is provided a fan device that includes a housing having a columnar shape and defining a central vertical axis. The housing includes an inlet configured to allow air into the housing, and an outlet formed around a circumference of the housing. There is a fan positioned inside the housing between the inlet and the outlet, and it is configured to blow air toward the outlet. There is a directional ducting member positioned inside the housing, between the fan and the outlet, to direct air blown by the fan through the outlet, radially, around the circumference of the housing. The air flow in all directions is continuous. The outlet is formed as a gap between the housing and the directional ducting member at the top of the housing. In accordance with a further feature, the housing has a circular hyperboloid shape.

[0008] In accordance with a further feature, wherein the directional ducting member has a circular hyperboloid shape, and wherein a distance between an inside of the housing and the directional ducting member decreases between the fan and the outlet.

[0009] In accordance with a further feature, the inlet is positioned above a base of the fan device, and the outlet is positioned at a top of the comfort fan.
[0010] In accordance with a further feature, the outlet is configured to direct air at an angle of between five and fifteen degrees relative to horizontal.
[0011] In accordance with a further feature, the fan is oriented to blow air in a vertical direction.
[0012] In accordance with a further feature, there is further included a control which is operable to adjust an air flow rate by controlling a speed of the fan.
[0013] In accordance with a further feature, the control includes a manual control provided on the fan device.
[0014] In accordance with a further feature, the control further includes a wireless transceiver operable to receive control information that indicates a speed of the fan to control the air flow rate.
[0015] In accordance with a further feature, there is further provided a temperature sensor, wherein an ambient temperature indicated by the temperature sensor is used, at least in part, by the fan device to set an initial air flow rate out of the outlet.
[0016] In accordance with a further feature, there is further included a humidity sensor, and wherein an ambient relative humidity indicated by the humidity sensor is further used by the fan device to set an initial air flow rate out of the outlet.
[0017] In accordance with a further feature, there is further included a light element that can be selectively illuminated to provide light outward from the fan device.
[0018] In accordance with the inventive embodiments described herein, there is provided a comfort fan that includes a housing having a vertical axis, an air inlet and an air outlet. The air outlet is configured to direct air out of the housing, radially to the vertical axis, around a three hundred sixty-degree circumference of the comfort fan, continuously. There are fan blades of a fan positioned inside the housing between the air inlet and the air outlet. The continuous radial airflow is accomplished without any oscillation of the comfort fan, and only the fan blades inside the comfort fan move; there are no other moving parts. There is a motor coupled to the fan blades to drive the fan blades at a selected speed. There is also a directional ducting member, having a circular hyperboloid shape, positioned inside the housing between the fan blades and the air outlet. The directional ducting member directs air blown by the fan blades through the air outlet radially outward, generally perpendicular to the vertical axis. A distance between the directional ducting member and the housing, inside the comfort fan, decreases from a lower terminus of the directional ducting member and the air outlet.
[0019] In accordance with a further feature, the air outlet is configured to direct air out of the housing radially entirely around the vertical axis, throughout an angle of three hundred sixty degrees.
[0020] In accordance with a further feature, the air outlet is positioned above the air inlet.
[0021] In accordance with a further feature, the air outlet is positioned at a height of eight to fifteen inches above a bottom of the comfort fan.
[0022] In accordance with a further feature, the motor is operated to provide an air flow rate of 160 to 400 feet minute at a selected distance from the comfort fan.
[0023] In accordance with a further feature, there is further included a temperature sensor, wherein an ambient temperature indicated by the temperature sensor is used, at least in part, to set an initial air flow rate out of the air outlet.
[0024] In accordance with a further feature, there is further included a humidity sensor, and wherein an ambient relative humidity indicated by the humidity sensor is further used by the comfort fan to set an initial air flow rate out of the outlet.
[0025] In accordance with a further feature, there is further included a light element that can be selectively illuminated to provide light outward from the comfort fan.
[0026] Although the invention is illustrated and described herein as embodied in a comfort fan device that provide continuous radial air flow, it is, nevertheless, not intended to be limited to the

details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

[0027] Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. The figures of the drawings are not drawn to scale.

[0028] Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The term “radius” as used herein is defined as the distance from a locus or point to a position on a curve around that locus. The terms “a” or “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically. The term “providing” is defined herein in its broadest sense, e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time.

[0029] “In the description of the embodiments of the present invention, unless otherwise specified, azimuth or positional relationships indicated by terms such as “up”, “down”, “left”, “right”, “inside”, “outside”, “front”, “back”, “head”, “tail” and so on, are azimuth or positional relationships based on the drawings, which are only to facilitate description of the embodiments of the present invention and simplify the description, but not to indicate or imply that the devices or components must have a specific azimuth, or be constructed or operated in the specific azimuth, which thus cannot be understood as a limitation to the embodiments of the present invention. The terms “vertical” and “horizontal” have their ordinary meaning; vertical is in a direction parallel to the gravitational vector, and horizontal is at a right angle to vertical. When used to describe components or structural elements of the inventive embodiments it shall be assumed that the device, machine, or system is oriented as intended for use as described herein. Furthermore, terms such as “first”, “second”, “third” and so on are only used for descriptive purposes, and cannot be construed as indicating or implying relative importance.

[0030] In the description of the embodiments of the present invention, it should be noted that, unless otherwise clearly defined and limited, terms such as “installed”, “coupled”, “connected” should be broadly interpreted, for example, it may be fixedly connected, or may be detachably connected, or integrally connected; it may be mechanically connected, or may be electrically connected; it may be directly connected, or may be indirectly connected via an intermediate medium. As used herein, the terms “about” or “approximately” apply to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure. In this document, the term “longitudinal” should be understood to mean in a

direction corresponding to an elongated direction of the article being referenced. The terms "program," "software application," and the like as used herein, are defined as a sequence of instructions designed for execution on a computer system. A "program," "computer program," or "software application" may include a subroutine, a function, a procedure, an object method, an object implementation, an executable application, an applet, a servlet, a source code, an object code, a shared library/dynamic load library and/or other sequence of instructions designed for execution on a computer system. Those skilled in the art can understand the specific meanings of the above-mentioned terms in the embodiments of the present invention according to the specific circumstances.
[0031] As used herein, the term "hyperboloid" refers to a geometric shape, sometimes called a hyperboloid of revolution or a circular hyperboloid, that is formed by rotating a hyperbola around an axis. Generally, herein, the axis will be a vertical axis unless otherwise stated. The hyperboloid is the surface obtained from a hyperboloid of revolution by deforming it by means of directional scalings, or more generally, of an affine transformation.
[0032] Conjunctive language such as the phrase "at least one of X, Y, and Z," unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0033] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present invention.
[0034] FIG. **1** is an exploded perspective view of a comfort fan that provides continuous radial air flow, in accordance with some embodiments.
[0035] FIG. **2** is a side elevational view of a comfort fan that provides continuous radial air flow, in accordance with some embodiments.
[0036] FIG. **3** is a perspective cut-away view of a comfort fan that provides a continuous radial air flow, showing the internal components of the comfort fan, in accordance with some embodiments.
[0037] FIG. **4** is a perspective cut-away view of a comfort fan that provides a continuous radial air flow, with most of the internal components of the comfort fan removed to show the internal air volume, in accordance with some embodiments.
[0038] FIG. **5** shows a side perspective view of a comfort fan that provides a continuous radial air flow showing an alternative arrangement of internal components, in accordance with some embodiments.
[0039] FIG. **6** shows a side profile view of a comfort fan that provides a continuous radial air flow, particularly showing the outward air flow and the elevation of the air outward air flow over a surface on which the comfort fan is located, in accordance with some embodiments.
[0040] FIG. **7** shows a top plan view of a use application of a comfort fan that provides a continuous radial air flow with the comfort fan placed in the middle of a round table, in accordance with some embodiments.
[0041] FIG. **8** shows an alternative configuration of a comfort fan that provides a continuous radial air flow, in accordance with some embodiments.
[0042] FIG. **9** shows a block schematic diagram of a comfort fan that provides a continuous radial

air flow, in accordance with some embodiments.

[0043] FIG. **10** shows a system diagram of a comfort fan that provides a continuous radial air flow that is operated in conjunction with a smartphone application program to automate operation of the comfort fan, in accordance with some embodiments.

[0044] FIG. **11**A shows an overhead view of a prior art oscillating fan.

[0045] FIG. **11**B shows overhead views of a prior art oscillating fan throughout its range of movement.

[0046] FIG. **12** shows a perspective view of a comfort fan having a hyperboloid shape, in accordance with some embodiments.

[0047] FIG. **13** shows a cross sectioned view of the comfort fan of FIG. **12**, in which the major structural components are shown.

[0048] FIG. **14** shows a graph chart plotting air speed of a fan versus ambient temperature, indicating a zone of comfort in which the air speed incident on a person is sufficient to produce a desirable cooling effect.

DETAILED DESCRIPTION

[0049] FIG. **11**A shows an overhead view of a prior art oscillating fan **1100**, which includes a cage **1102** in which a set of fan blades **1104** with multiple blades is located. The cage **1102** prevents inadvertent contact with the spinning fan blades **1104** during operation of the fan **1100**. The fan blades **1104** are driven by a motor **1106**. The motor **1106** is mounted on a column or other support that is attached, at its lower end, to a base that rests on the floor or other surface. The main portions of the fan **1100**, including the cage **1102**, blades **1104**, and motor, oscillate around the support as driven by the motor **1106** through a gear or cam structure, as indicated by arrows **1110**, as is well known. The blades are oriented to blow air generally in a horizontal direction, as indicated by arrow **1108**, when the fan **1100** is placed on a horizontal surface. If the fan **1100** does not oscillate, then the main air flow from the fan **1100** is experienced only in a narrow angle directly in front of the fan. Of course, in an enclosed room, air throughout the room will be moved, but in, for example, an outdoor environment, if the fan **1100** is not oscillating, a person would have to be directly in front of the fan **1100** to receive air flow. FIG. **11**B shows what happens when the fan **1100** is set to oscillate. In general, the fan **1100** can oscillate throughout a range of motion, from a first end position **1112** to a second end position **1116**, passing through all positions, including middle position **1114** throughout each oscillation cycle. While this allows a broader range of air flow, the air flow at any given fixed position in the range of oscillator will be only momentary as the fan oscillates continuously. Prior art oscillating fans only oscillate through a range that is typically not more than about one hundred twenty degrees. Thus, to provide airflow to several people, for example, around a table, the fan would have to be moved away from the table so that all people at the table are within the oscillation range of the fan. But then, some people will be closer to the oscillating fan and others further away from the oscillating fan, and as a result, they will experience not only monetary air flow, but substantially different rates of air flow. In an indoor environment, a ceiling fan can be mounted over a seating area, but in an outdoor environment, there is no ceiling to which a ceiling fan can be mounted. Thus, a solution to providing a comforting air flow to people around an area like a table is needed.

[0050] While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. It is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms.

[0051] The present invention provides a novel and efficient comfort fan that provides a continuous radial air flow at a calculated velocity that is selected to provide a cooling/heating effect and thermal satisfaction based on ambient conditions and personal factors such as metabolic rate and clothing insulation level. The comfort fan draws air in and directs it outward, in a radial manner, at a velocity which is selected such that at the distance a person is located from the fan outlet, the incident air flow will provide an optimal cooling or warming effect. The flow rate can be adjusted to personal preference, but is initially based in a range of air velocity between that which is considered to be minimally effective (160 feet per minute) and a velocity considered to be a nuisance (more than 400 feet per minute). The radial out-flow of air is provided from an elevated position above a surface, such as a table, to avoid blowing air over items on the surface, such as, for example, food items, and the radial air flow can be aimed slightly upwards so that the air flow is incident on people around the fan at about the chest, neck, and face level. The comfort fan avoids the intermittency problem of oscillating fans by providing a continuous air flow radially from the fan, and is ideal for setting up in the center of a table in a warm setting (e.g. outdoors) and providing a gentle cooling air flow to all those seated around the table. Embodiments of the invention provide for warming air flows as well by including a heating element inside the comfort fan. In addition, the comfort fan can include lighting elements to provide lighting for mood as well as task lighting, as desired.

[0052] FIG. **1** is an exploded perspective view of a comfort fan **100** that provides continuous radial air flow, in accordance with some embodiments. FIG. **2** shows a side elevational view of the comfort fan **100**, and reference should be made to both drawings. The comfort fan **100** in general has a generally columnar shape with a central vertical axis **210**, and draws in ambient air through an inlet section **110**, as indicated by arrow **206**, and blows the air out through an outlet formed between the top of an upper housing section **114**. It will be appreciated by those skilled in the art that the precise shape of the fan, and specifically the housing section **114**, while being generally columnar, can vary. This means, for example, that the housing may not have a uniform horizontal diameter along its height and may follow any of a variety of curves or shapes. In some embodiments, the housing section **114** can have a circular hyperboloid shape in which the top flares outward and the side, in the vertical direction, follows a hyperbolic or inverted parabolic curve.

[0053] The air blown by the comfort fan **100** is directed, at least in part, by a directional ducting member, such as a conic director **116**. The conic director **116** can have a variety of shapes, including straight or linear conic, circular hyperboloid, or any other shape in which the horizontal diameter increases along its vertical axis from bottom to top. The outlet air flow is represented by arrow **208**, and exits the fan **100** radially around the fan. Said another way, the outlet is formed radially around the fan **100** to generally direct air outward in a horizontal direction, around a substantial angle around the fan **100**. In the present example, the fan **100** provides air flow around the entire fan **100**, meaning an angle of three hundred sixty degrees. The air flow speed can be moderated or adjusted so that the air flow incident on people who are a distance from the fan **100** is at an optimal rate to provide a desired comfort effect.

[0054] Thus, as used herein, it should be understood that the term “radial” in reference to air flow from the comfort fan **100** means continuous air flow in a substantial angle around the fan **100**. In some embodiments the radial air flow is circumferential, meaning that air is blown out from the comfort fan around the entire circumference of the comfort fan. The outlet is a circular outlet around, or at least partially around, a central vertical axis **210** defined by the comfort fan **100**. Being columnar, the comfort fan **100** is taller (vertical direction) that it is wide (horizontal direction), and, as described, can have a variety of shapes. The comfort fan produces a continuous outward air flow through the entire angle of the circular outlet, and as a result, any persons around the comfort fan will receive a continuous flow of air, even people positioned on opposite sides of the comfort fan from each other. This eliminates the effect of an oscillating fan, which only blows air in a given direction periodically and momentarily. It is contemplated that the circular outlet can be less than a complete circle, and instead follow a circular path around the fan that is less than a full circle. This can allow, for example, a comfort fan that is intended to be placed against a wall, on a table surface, and provide an outflow of air around an angle of one hundred eighty degrees instead of a complete circle around the fan, since blowing air into the wall would be inefficient.

[0055] The fan **100** includes a base **102** that can house some or all of the control electronics, a battery, and other components. In some embodiments the base **102** can be removable to allow recharging of the battery without having to move the entire fan **100**. In that way bases or batteries can be easily swapped out as one battery becomes depleted of charge and needs to be recharged. A lower separator **104** separates the base **102** from the air inlet chamber, preventing any dust or other particular material from accumulating in the base **102**. A parabolic air director **106** can sit on top of the separator **104**, inside of the housing inlet section **110**. The housing inlet section **110** comprises openings **124** (e.g. slits or slots) though which air is drawn into the fan **100**. The air director **106** can be open at the top/middle to accommodate a motor **108** that drives fan blades **112**. The motor **108** is oriented so that its drive shaft is vertical. In some embodiments, however, the fan blades **112** can be rim-driven by a shaftless motor. The fan blades **112** are turned at a selected rate to provide the continuous radial air flow at a rate that provides the desired cooling/heating effect and thermal comfort. The air director **106** helps provide a laminar flow of air to the fan blades **112** in a manner that reduces the sound generated by the fan blades **112** that would otherwise occur without the air director **106** being present and instead having a fully open volume below the fan blades **112**. The lower end of the upper housing section **114** surrounds the fan blades **112**, leaving only a small gap between the inner surface of the upper housing section **114** and the fan blades **112** to optimize the efficiency of moving air from the inlet section **110** into the upper housing section **114**. A generally funnel-shaped conic director **116** sits in the upper housing section **114**, and has a lower apex **130** that sits directly over the center of the fan blades **112**. There can be a plurality of rib walls or vanes **126** that extend from the surface **128** of the conic director **116** that bear against the inner surface **132** of the upper housing section **116**, resulting in a space between the surface **128** of the conic director **116** and the inner surface **132** of the upper housing section **114** through which air flows from the fan blades **112**. In other embodiments, posts can be used to reduce turbulence that may result from vanes **126**. The conic director **116** can have any of a variety of shapes that are generally conic, including paraboloid and hyperboloid shapes. In general, the lower terminus **130** of the conic director **116** has a minimal diameter, and the diameter of the conic director increases with height between the lower terminus **130** and where the surface **128** meets the flange **134**. The top flange **134** at the top of the conic director **116** extends outward in the horizontal direction, blocking air flow in the vertical direction and directing air flow in the horizontal direction as indicated by arrow **208**. A cap **118** can fit over the top of the conic director **116**. To further help direct air in the horizontal direction, the diameter **202** of the top of the upper housing section **114** can be smaller than the diameter **204** of the flange **134**. In some embodiments the diameter **204** can be about **20**% larger than diameter **202**. The fan blades **112** move air upwards, in the present exemplary configuration, into the volume surrounded by the upper housing section **114**, and between the conic director **116** and the upper housing section **114**, forcing the air outward, radially, around the fan **100**, in a continuous air flow. In some embodiments the fan **100** can have a motor speed control **120** the allows a user to adjust the speed of the motor **108**/fan blades **112** to adjust the rate of air flow that is incident on the user. As shown here, inlet section **110** directs the air in horizontally, which is then turned vertically. In some embodiments, however, the air flow into the comfort fan **100** can be entirely vertical, which can improve flow velocity in some embodiments.

[0056] FIG. **3** is a perspective cut-away view of a comfort fan **300** that provides a continuous radial air flow, showing the internal components of the comfort fan, in accordance with some embodiments. FIG. **4** show the same view as FIG. **3** but with some of the internal components of the comfort fan **300** removed to show the internal air volume. The comfort fan **300** exemplifies another configuration of a comfort fan that provides a continuous radial air flow. A base section **302** provides the bottom of the housing and can hold a battery **308** and electronic/electrical elements for controlling operation of the comfort fan **300**. An inlet section **304** of the housing surrounds a lower internal volume **306**, and provides openings, such as slots or apertures through which air can be drawn into the lower internal volume **306**. A motor **310** can be vertically oriented in the center of the housing to drive fan blades **314** positioned in the lower end of the upper housing section **312**. The fan blades **314** can include a central cylindrical opening **316** into which the lower terminus **320** of the conic director **318** can sit such that the top of the blades of the fan blades **314** at about even with (vertically), or higher than the lower terminus **320**. The conic director **318** in this example has a hyperboloid shape, which can be considered a circular hyperboloid air duct, up to a top edge **324** that is about horizontally coextensive with the upper lip **322** of the upper housing section **312**. As a result, air blown by the fan blades **314** exits the comfort fan **100** at the gap between the top edge **324** and upper lip **322** generally in the horizontal direction and indicted by arrow **402**. Posts **323** can be used to maintain separation between the conic director **318** and the inner surface **325** of the upper housing section **312**. The posts **323** can be positioned away from the opening between the upper lip **322** and top edge **324** so as to conceal them from view. Air is drawn in through the inlet section **304** as indicated by arrows **406**. A circular circuit board **404** can surround inlet volume, and hold light elements such as light emitting diodes **408**. Light emitted from these LEDs **408** will illuminate the interior volume of the comfort fan, and light will be emitted out of the inlet **304** as well as the outlet between the upper edge **324** and upper lip **322**.

[0057] FIG. **5** shows a side perspective view of a comfort fan **500** that provides a continuous radial air flow showing an alternative arrangement of internal components, in accordance with some embodiments. The inlet section **502** sit directly on several feet **504**. The upper housing section **506** sits on the top of the inlet section **502**, and can be transparent. Fan blades **508** are positioned at the bottom or lower end of the upper housing section **506**. A conic director **510** is positioned above the fan blades **508** with the tip **518** of the conic director **510** centered over the fan blades **508**. Air is blown by the fan blades **508** upwards, and the air is redirected by the conic director **510** outward in a generally horizontal direction, radially around he fan **500**. The air blown by the fan blades **508** escapes out of an outlet formed by the gap between the outer edge **520** of the top **512** of the conic director **510**, which has a hyperboloid shape, and as a result, the profile of the surface **514** of the conic director, from the tip **518** to outer edge **520** is concave. The outer edge **520** of the top **512** of the conic director **510** extends outward farther than the top lip **522** of the upper housing section **506**. As a result, air is forced out between the conic director **510** and the upper lip **522** of the upper housing section **506**. The further help direct the air outward in the horizontal direction, the upper portion of the upper housing section **506** can also have a hyperboloid shape which complements that of the conic director **510**. Several standoffs **516** can be used to connect the conic director **510** to the top of the upper housing section **506**. The standoffs **516** can be cylindrical, having a common length, and with a threaded bore at each end to receive a screw that passes through the upper lip **522** or the outer edge **520**, with each standoff **516** holding a screw through both the conic director **510** at the outer edge **520** and the upper housing section **506** at the upper lip **522**. Thus, air is drawn in through the inlet **502**, moved by the fan blades **508** toward the conic director **510**, and exits the fan **500** radially in a circular pattern in the horizontal direction. The speed of the fan blades **508** can be adjusted such that the air flow rate incident on a user or other persons around the fan is at a desired comfort level.

[0058] FIG. **6** shows a side profile view of a comfort fan **602** that provides a continuous radial air flow, particularly showing the outward air flow and the elevation of the air outward air flow over a surface **600** on which the comfort fan **602** is located, in accordance with some embodiments. The surface **600** can be, for example, a table such as those commonly found in outdoor environments for dining or other leisure activity. The comfort fan **602** in the present example can be consistent with any of the exemplary embodiments disclosed herein, and those consistent with the exemplary embodiments disclosed herein.

[0059] In general, the comfort fan **602** is sized to sit on a surface, such as a table, that can be surrounded by several people (seated). For example, the surface **600** can be a circular patio or other outdoor-type table. Furthermore, the outlet of the comfort fan has a height **610** above the surface

600 such that the air is blown outward, in a radial horizontal circular pattern perpendicular to the central vertical axis **612**, or at a slight angle **614** to horizontal **616**, and is such that the air flow is not substantially incident on items on the surface **600**. The height **610** can be, in some embodiments, eight to fifteen inches. In other embodiments the height **610** can be more than fifteen inches or less than eight inches, as the particular application may require. Elevating the outlet of the comfort fan **602** over the surface **600** prevents the air flow from the comfort fan **602** from, for example, cooling hot food, blowing napkins or other items on the surface **600**, and so on. Rather, the air outflow is directed towards the people around the comfort fan **602**, whose torsos, head, and faces will be substantially elevated relative to the surface **600**. The angle **614** can be in the range of five to fifteen degrees in some embodiments. In some embodiments it has been found that an angle **614** of ten degrees is optimum for an ordinary circular table sized to seat four to six people. In other embodiments an angle larger than fifteen degrees or less than five degrees can be used, depending on the particular application. Generally, since the outlet of the comfort fan **602** will generally be at height that is lower than the chest/neck/face areas of persons seated around comfort fan **602**, a slight upward direction will ensure that the air flow is incident on the chest/neck/face areas of the people.

[0060] Air is drawn into the fan through an inlet or inlet section **608**, and is blown outward, radially in a circular pattern, horizontally as indicated by air flows **604**. The speed of the air flow is adjusted so that, at some distance **606** from the fan, which is where a person or persons are likely to be positioned relative to the comfort fan **602**, the air flow provides a desired cooling effect and thermal comfort. For example, it has been found that an air flow rate of about two hundred to four hundred feet per minute, over a region that generally spans from the chest to the top of a person's head, and adjustable to a speed in that range (or further outside of that range), will provide a desired thermal comfort effect. However, controls on the comfort fan **602** will allow people to adjust the air flow to a preferred air flow rate if the initial air flow rate, calculated based on temperature, and humidity conditions in some cases, is not preferable. That it, the initial air flow rate will depend on factors such as personal preference, activity level, humidity, operative (dry bulb and mean radiant) temperature, and clothing worn by the people around the comfort fan **602**. The flow rate can be measured at a generally standard distance from the center of a table configured to seat four to six peoples, for example. In some embodiments, the comfort fan **602** can include a temperature sensor and a humidity sensor to sense the ambient conditions, and based on an average of how far away from the comfort fan **602** people will be assumed to be positioned, and the ambient conditions, the comfort fan can select an initial fan blade speed to hopefully provide a desired comfort effect to the people actually around the comfort fan **602**. However, as individual preferences can vary, and assumptions may not always be precisely correct, the comfort fan **602** also includes controls to adjust the fan blade speed and the air flow rate. Under a set of given conditions, the person or people around the comfort fan may desire more or less air flow, and adjust the flow rate from the initial air flow rate to one that is more preferable.

[0061] As is well known, the incident flow rate of ambient temperature air can have a perceived cooling effect. It is also known that, at a certain flow rate and above, a person can perceive the air flow rate to be a nuisance by, for example, blowing the person's hair or even clothing excessively. Thus, the air flow rate that is incident on a person should be within a range that is perceived as having a desirable cooling effect, but not so high that it becomes an annoyance. Although, being adjustable, it is contemplated that a given person may preferable a flow rate that would be considered insufficient, or a nuisance rate by other people. The effect of moving air under different environmental conditions has been studied in great detail. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), for example, have produced their Standard 55-2023, titled Thermal Environmental Conditions for Human Occupancy, which describes how environmental factors (temperature, thermal radiation, humidity, and air speed) and personal factors (activity and clothing) combine in complex ways to create thermal conditions, and how personal comfort levels vary from one occupant to another. The Standard is designed to specify those combinations of factors that result in satisfactory thermal conditions for a majority of occupants. FIG. **14** discusses this standard and the tool developed by the Center for the Built Environment to identify air flow rates based on environmental factors in the discussion below.

[0062] FIG. **7** shows a top plan view of a use application of a comfort fan **704** that provides a continuous radial air flow with the comfort fan **704** placed in the middle of a round table **702**, in accordance with some embodiments. The comfort fan **702** in the present example can be consistent with any of the exemplary embodiments of comfort fans disclosed herein, and those consistent with the exemplary embodiments disclosed herein. The comfort fan **704** is placed at the center of the table **702**. Around the table **702** are several chairs **710** in which people can sit. Since the table **702** is circular, each of the chairs **710** are about the same distance from the center of the table. A dashed circle **706** represents a radial distance **708** from the comfort fan **704** at which people will be seated, on average, given the diameter of the table **702**, and assuming the comfort fan **704** is placed in the center of the table **702**. The radial distance **708** can be a selected distance, either as an assumption for common tables, or as an input provided by a user of the comfort fan **704**. Air is blown radially, in a continuous manner, from the comfort fan, as indicated by arrows **710**. The initial flow rate of the air can be based on ambient conditions (e.g. operative temperature and humidity), and the radial distance **708** at which people are generally seated from the comfort fan **704**. The initial air flow rate is generated by the initial fan speed, which can be a selected speed based on the environmental factors, the relationship between fan speed and air velocity at the outlet of the comfort fan, and the selected distance. The comfort fan **704** can also include manual controls that, among other operations, allow a person to adjust the fan blade speed so that the incident air flow rate at the positions where people are seated around the comfort fan **704** is at a preferred rate.

[0063] In some embodiments, the comfort fan **704** can include a wireless radio transceiver, such as a transceiver designed in accordance with the BLUETOOTH wireless networking protocol, which allows the comfort fan **704** to wirelessly connect to, for example, a smartphone device that is running an application program designed to allow a person to control the comfort fan **704** via the smartphone device. The smartphone device, by executing the application program, can receive input regarding the air flow rate (e.g. to increase or decrease the air flow rate). Other input can be received as well, such as activating a heating element in the comfort fan to provide warm air. Furthermore, it is contemplated that the comfort fan **704** can also include lighting elements to provide light of various colors, including changing color over time, as well as light that can enable persons around the comfort fan **704** to read and see materials placed on the table **702** when the ambient lighting is dark. The comfort fan **704** generates a radial airflow around the circumference of the comfort fan **704**, continuously around the entire three hundred sixty-degree circumference of the comfort fan **704**. In addition to radial air flow, as indicated by arrows **712**, light can be provided by the comfort fan **704** similarly, outward from the comfort fan **704** radially.

[0064] FIG. **8** shows an alternative configuration of a comfort fan **800** that provides a continuous radial air flow, in accordance with some embodiments. The comfort fan **800** can be useful in situations where the people around the comfort fan **800** are seated substantially closer to the comfort fan **800** than in the examples of FIGS. **6-7**. The comfort fan **800** is shown here partially in cut-away to expose some of the internal components. The comfort fan **800** includes a base **802** which can house control components, and/or a battery or electrical connector, or both. A conic director **806** is inverted relative to the conic directors shown in embodiments discussed previously, and includes an attachment section **804** that can connect with the base **802**. A middle housing section **808** can provide an outlet through a plurality of slots. Air blown by the fan blades **810** is drawn in through in inlet **812**, as indicated by arrow **814** and drawn over the conic director **806** (which has a hyperboloid shape), and out of the outlet **818**, as indicated by arrow **816**. The comfort fan **800** can be used in more intimate settings, where, for example, a smaller fan can be used compared to that of a larger table used, for example, for dining. Overall, the comfort fan **800** has a cylindrical shape with a circular horizontal cross section. The air outlet **818** is configured to direct air outward, horizontally and radially in all directions to provide continuous air flow to people gathered round the comfort fan **800**. Since the comfort fan **800** does not oscillate, the air flow is simultaneously experienced at any point around the comfort fan **800**. The comfort fan **800** can also include controls that can be adjusted manually, or wirelessly (e.g. via a phone application program).

[0065] FIG. **9** shows a block schematic diagram of a comfort fan **900** that provides a continuous radial air flow, in accordance with some embodiments. The electrical and electronic components represented herein can be consistent with any of the exemplary comfort fan embodiments disclosed herein, and those other equivalent embodiments consistent with the exemplary embodiments disclosed herein. The comfort fan **900** can include a controller **902**, such as a microcontroller or microprocessor that is capable of performing instruction code that is designed to, for example, control fan speed in relation to various input, control lighting and heating elements as well, among other functions. The controller **902** can be coupled to a memory **904** which can represent an aggregate memory that includes both long term storage memory (e.g. read only memory) as well as “scratchpad” memory (e.g. random access memory). The controller **902** can further be coupled to manual controls **906** such as one or more potentiometers for adjusting various fan operations, such as fan blade speed. The resistance of the potentiometer can be sensed by a voltage divider and the voltage can be converted into a digital value by an analog to digital converter port of the controller **902**. The controller **902** can further be coupled to a fan speed module **908** that controls the speed of the fan blades of fan **910**. In some embodiments the comfort fan **900** can include a temperature sensor **912** that senses the ambient temperature. Further, a humidity sensor **918** can also be used in conjunction with temperature information so that an initial fan speed can be selected, which can then be adjusted, if necessary, using one of the controls **906**. The temperature sensor **912** indicates the present ambient temperature, and the humidity sensor indicates the present ambient relative humidity. In some embodiments temperature alone can be used to set the initial fan speed, and in some embodiments both temperature and relative humidity are used to calculate an appropriate initial fan speed.

[0066] In some embodiments, the comfort fan **900** can also include lighting elements **914** that can be adjusted using one or more of the controls **906**. For example, a color selection can be made, as well as a brightness level. In some embodiments, the light color can change over time, gradually changing from one color to another. Furthermore, in some embodiments a heating element **909** can be present to heat the air drawn into the comfort fan **900** in cases where the ambient temperature is lower than a comfort threshold so that warm air is blown out to the people around the fan **900**. Finally, in some embodiments, there can be a wireless transceiver **916** that allows the fan **900** to wirelessly connect to, for example, a smartphone or similar computing device. It is contemplated that a smartphone (or equivalent) device can run an application program designed to control the comfort fan, providing both initial control as well as input for manual control of the comfort fan **900** as well. Control information can be transmitted to the wireless transceiver **916** that indicates a fan speed, or fan speed adjustment information so that a user can adjust the fan speed to a more preferable speed if the initial fan speed is not preferable.

[0067] FIG. **10** shows a system diagram **1000** of a comfort fan **1002** that provides a continuous radial air flow that is operated in conjunction with a smartphone application program to automate operation of the comfort fan, in accordance with some embodiments. The fan **1002** can be consistent with any of the exemplary comfort fan embodiments disclosed herein, and those other equivalent embodiments consistent with the exemplary embodiments disclosed herein. The fan **1002** can be set on a surface **1004**, such as a table, counter, or other surface near people. The fan **1002** is intended to provide an air flow or either ambient air, if the ambient air temperature is warm, or heated air if the ambient air temperature is cool/cold. The fan **1002** can be wirelessly coupled to a computing device, such as a smartphone device **1006** that runs an application program designed to allow a user to control operation of the fan **1002**. For example, the application program can provide graphical elements to receive input for controlling fan speed and lighting elements of the fan **1002**. In some embodiments the smartphone can determine its location while connected to the fan using, for example, satellite positioning signals from overhead positioning satellites **1008**. The location can then be cross referenced with a weather report for the region indicating the present temperature, which the smartphone device, via the application program, can use to set an initial fan speed that can be subsequently adjusted by the user, if so desired. In some embodiments, if the weather information indicates that the ambient temperature is cold, then the smartphone device **1006** can activate a heating element in the fan **1002** to provide warm air.

[0068] FIG. **12** shows a perspective view of a comfort fan **1200** having a hyperboloid shape, in accordance with some embodiments. In general, the comfort fan **1200** is columnar with a hyperboloid shape to the main body. FIG. **13** shows a cross sectioned view of the comfort fan, with the section being vertical along line B-B′. Reference should be made to both drawings for the following discussion. They hyperboloid shape is similar to that of the comfort fan embodiments shown in FIGS. **2-6**. The comfort fan includes a base **1202** in which there can be, for example, electronic to operate the comfort fan **1200**, light elements, a user interface. A battery and/or power circuitry can also be included in the base **1202**. The base **1202** is generally circular when viewed from the top or bottom (i.e. in a horizontal plane), and can include a user interface section **1224** that extends outward from the base **1202**, and can include user interface elements to control and operate the comfort fan **1200**, including, for example, a graphic display screen and buttons.

[0069] An inlet air director **1204** sits over the base and the surface of the inlet air director **1204** has a hyperboloid shape from the outside edge **1212** up to a top **1306** of the inlet air director **1204**. The hyperboloid shape of the inlet air director **1204** is such that the top **1306** is much smaller in diameter than the outer edge **1212**, follows a hyperbolic curve in a direction from the outer edge **1212** to the top **1306**, and the top **1306** of the inlet air director **1204** terminates just below an axial fan motor **1308**. The comfort fan **1200** also includes two outer side housing portions, including a lower side housing portion **1206** and an upper side housing portion **1216**. The side housing portions **1206**, **1216** have an outer surface that, in the vertical direction, follows a hyperboloid curve. In particular, the bottom **1210** of the lower side housing portion **1206** extends outward farther than the top of the lower side housing portion **1206** where it meets the bottom of the upper side housing portion **1216**. In some embodiments, where the side housing portions **1206**, **1216** meet can be the narrowest point, in the horizontal direction, of the side housing portions **1206**, **1216**. Between the bottom edge **1210** of the lower side housing portion **1206** and the outer edge **1212** of the base **1202** is a circumferential gap **1208** that acts as an air inlet.

[0070] The upper side housing portion **1216** continues the hyperbolic curve outward to a top edge **1226**. There is a circumferential gap **1220** between the top edge **1226** and the outer edge **1218** of the upper air director **1222**. The upper air director **1222** and the lower air director **1204** are both directional ducting members as they direct air through the interior of the comfort fan, from the inlet **1208**, through the fan, and out of the outlet **1220**. The upper air director **1222** has a hyperboloid shape along its surface **1318**, being narrowest, in the horizontal direction, at a lower tip **1316** that is positioned directly over the axial fan and centered on the axis **1315** of the axial fan. In the present embodiment, the outer or outside surface of the upper air director **1222** can follow the inner surface **1318**, resulting in a hyperboloid cavity at the top of the comfort fan **1200**. The upper air director **1222** can be mounted to the inner side **1322** of the upper side housing portion **1216** by several posts **1320** that extend vertically from the inner surface **1322** of the upper side housing portion to the inner surface **1318** of the upper air director **1222**. The material of the upper air director **1222** can be heat staked, screwed, glued, or otherwise affixed to the post **1320** at each instance of the posts **1320**.

[0071] The lower side housing portion **1206** has a top portion **1310** that turns inward, and follows the circumference of top of the lower housing portion **1206**. Similarly, a bottom portion **1312** of the upper side housing portion **1216** abuts the top portion **1310**, and also turns inward, thereby creating a region where the top portion **1310** can be joined or fastened to the bottom portion **1312**. However, the bottom portion **1312** of the upper side housing portion **1216** forms a cowl **1314**, which is a vertically oriented band around the circumference of an inside edge of the bottom portion **1312** of the upper side housing portion **1216**. The cowl **1314** acts as a duct, and the fan blades of the axial fan are positioned in the cowl **1314** to create a highly efficient movement of air through the comfort fan **1200**.

[0072] The lower side housing member **1206** can include three or more vertically oriented vanes **1214** that are planar in a radial direction from the central axis, which the vertically through the shaft **1315** of the axial fan. The bottoms **1304** of the each of the vanes **1214** can be captured in a respective slot formed between parallel raised runners **1302** along the surface of the lower air directors **1204**. This allows the housing assembly, comprising the lower side housing portion **1206**, the upper side housing portion **1216**, and upper air director **1222** to be easily lifted and removed from the base **1202** and lower air director **1204**. The housing assembly can be replaced by aligning the vanes **1214** with the slots formed by the parallel raised runners **1302**.

[0073] In operation, when the axial fan motor **1308** is turned on, the axial fan blades (disposed within the cowl **1314**, will turn in a direction that moves air from the inlet **1208**, through the interior of the comfort fan **1200**, and out of the air outlet **1220**, with the air movement being guided by the lower air director **1204**, the surface **1318** of the upper air director **1222**, and hyperboloid curve of the interior surfaces **1321**, **1322** of the lower and upper side housing portions **1206**, **1216**. Going upwards, the distance between the upper side housing portion **1216** and the upper air director **1222** decreases, and as a result, the pressure of the air moving through the air outlet **1220** is increased. For example, the distance **1326** between the surface **1318** of the upper air director **1222** and the inside if the upper housing portion **1216** is much larger than the distance **1328** near the air outlet **1220**. This narrowing of the outlet, despite the larger circumference, increases the air velocity at the outlet **1220** relative to that lower inside the comfort fan. The relationship between fan speed and air velocity at the outlet **1220** can be characterized, such as by testing, and used to determine the optimum fan speed for an initial air velocity calculated based on environmental factors. The outer edge **1218** of the upper air director **1222** extends outward in the horizontal direction farther than the top edge **1226** of the upper side housing portion **1216** by a distance **1324**, which is configured to direct air outward, radially (mostly in the horizontal direction), and circumferentially (i.e. in a 360° direction) around the comfort fan **1200**. The velocity of the air exiting the air outlet **1220** is controlled so that, at a given distance from the comfort fan the air will be moving at a rate that falls within a comfort region for the given ambient environmental conditions. That distance can be, for example, from the center of a round table to the seating positions around the table, as described in reference to FIG. **7**.

[0074] FIG. **14** shows a graph chart **1400** plotting air speed **1404**, in feet per minute, incident on a person, versus ambient temperature **1402**, indicating a zone **1406** of comfort in which the air speed incident on the person is sufficient to produce a desirable and substantially cooling effect. The chart **1400** was produced using the THERMAL COMFORT TOOL of the Center for the Built Environment, which implements Specification Standard 55-2023 of the ASHRAE, as provided at https://comfort.cbe.berkely.edu. The parameters used to generate the chart **1400** were selected to be similar to those of Miami, Florida. The operative temperature was set to 83.6 degrees F., and the relative humidity was set to 84.9%. An air speed of 200 feet per minute (fpm) is represented at location **1410** in the zone **1406**. Beyond 79 degrees F. the necessary air speed increases sharply, and at 83.6 degrees F. a minimum air speed of 160 fpm is needed at the individual's exposed skin to provide a sufficient cooling effect. In the present example a cooling effect of 5.3 degrees F. is achieved. The air speed of 200 fpm is measured at the individual. To generate that air speed at the individual, the comfort fan must output air at a greater speed since it is a given distance (e.g. distance **708** in FIG. **7**) from the individual. However, the calculation of output air speed from the comfort fan to achieve the desired air speed at the given distance from the fan is known, and can be programed into the comfort fan so that the controller (e.g. **902**) can control the fan speed to output air at a speed that achieves the desired thermal comfort effect at the given distance from the comfort fan.

[0075] The inventive comfort fan as disclosed herein avoids the problems associated with prior art fans, such as oscillating fans, that only intermittently provide air flow across a wide angle. The inventive comfort fan uses a radial ducting system to direct airflow outward from the comfort fan around a wide angle, if not completely around the comfort fan, continuously so that at any point within the radial field equidistant from the comfort fan the same flow rate of air will be experienced as a continuous flow. The comfort fan draws in air at, or near its base, and directs the air, by fan blades, to a conic director, such as a circular hyperboloid air duct, which distributes the air flow evenly outward in a circle around the comfort fan, or at least around a large angle (e.g. one hundred twenty degrees or more) that is not achievable with conventional fans.

[0076] The claims appended hereto are meant to cover all modifications and changes within the scope and spirit of the present invention.

Claims

1. A fan device, comprising: a housing having a columnar shape and defining a central vertical axis; an inlet configured to allow air into the housing; an outlet formed around a circumference of the housing; a fan positioned inside the housing between the inlet and the outlet, and configured to blow air toward the outlet; a directional ducting member positioned inside the housing between the fan and the outlet to direct air blown by the fan through the outlet radially around the circumference of the housing; and wherein the outlet is formed as a gap between the housing and the directional ducting member.

2. The fan device of claim 1, wherein the housing has a circular hyperboloid shape.

3. The fan device of claim 2, wherein the directional ducting member has a circular hyperboloid shape, and wherein a distance between an inside of the housing and the directional ducting member decreases between the fan and the outlet.

4. The fan device of claim 1, wherein the inlet is positioned above a base of the fan device, and the outlet is positioned at a top of the comfort fan.

5. The fan device of claim 1, wherein the outlet is configured to direct air at an angle of between five and fifteen degrees relative to horizontal.

6. The fan device of claim 1, wherein the fan is oriented to blow air in a vertical direction.

7. The fan device of claim 1, further comprising a control which is operable to adjust an air flow rate by controlling a speed of the fan.

8. The fan device of claim 7, wherein the control includes a manual control provided on the fan device.

9. The fan device of claim 7, wherein the control further includes a wireless transceiver operable to receive control information that indicates a speed of the fan to control the air flow rate.

10. The fan device of claim 1, further comprising a temperature sensor, wherein an ambient temperature indicated by the temperature sensor is used, at least in part, by the fan device to set an initial air flow rate out of the outlet.

11. The fan device of claim 10, further comprising a humidity sensor, and wherein an ambient relative humidity indicated by the humidity sensor is further used by the fan device to set an initial air flow rate out of the outlet.

12. The fan device of claim 1, further including a light element that can be selectively illuminated to provide light outward from the fan device.

13. A comfort fan, comprising: a housing having a vertical axis, an air inlet and an air outlet; the air outlet being configured to direct air out of the housing radially to the vertical axis around a three hundred sixty-degree circumference of the comfort fan continuously; fan blades positioned inside the housing between the air inlet and the air outlet; a motor coupled to the fan blades to drive the fan blades at a selected speed; and a directional ducting member having a circular hyperboloid shape positioned inside the housing between the fan blades and the air outlet to direct air blown by the fan blades through the air outlet radially outward perpendicular to the vertical axis, wherein a distance between the directional ducting member and the housing decreases from a lower terminus of the directional ducting member and the air outlet.

14. The comfort fan of claim 13, wherein the air outlet is configured to direct air out of the housing radially entirely around the vertical axis, throughout an angle of three hundred sixty degrees.

15. The comfort fan of claim 13, wherein the air outlet is positioned above the air inlet.

16. The comfort fan of claim 13, wherein the air outlet is positioned at a height of eight to fifteen inches above a bottom of the comfort fan.

17. The comfort fan of claim 13, wherein the motor is operated to provide an air flow rate of 160 to 400 feet minute at a selected distance from the comfort fan.

18. The comfort fan of claim 13, further comprising a temperature sensor, wherein an ambient temperature indicated by the temperature sensor is used, at least in part, to set an initial air flow rate out of the air outlet.

19. The comfort fan of claim 18, further comprising a humidity sensor, and wherein an ambient relative humidity indicated by the humidity sensor is further used by the comfort fan to set an initial air flow rate out of the outlet.

20. The comfort fan of claim 13, further including a light element that can be selectively illuminated to provide light outward from the comfort fan.
